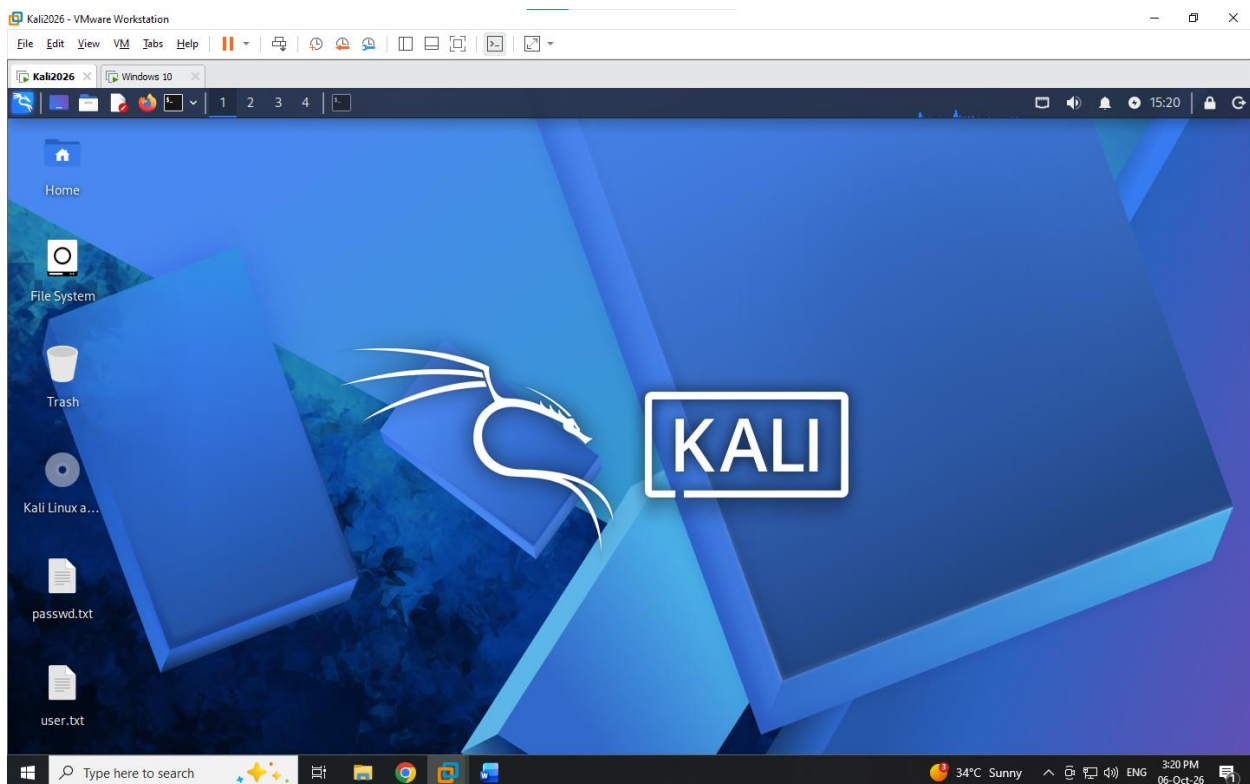
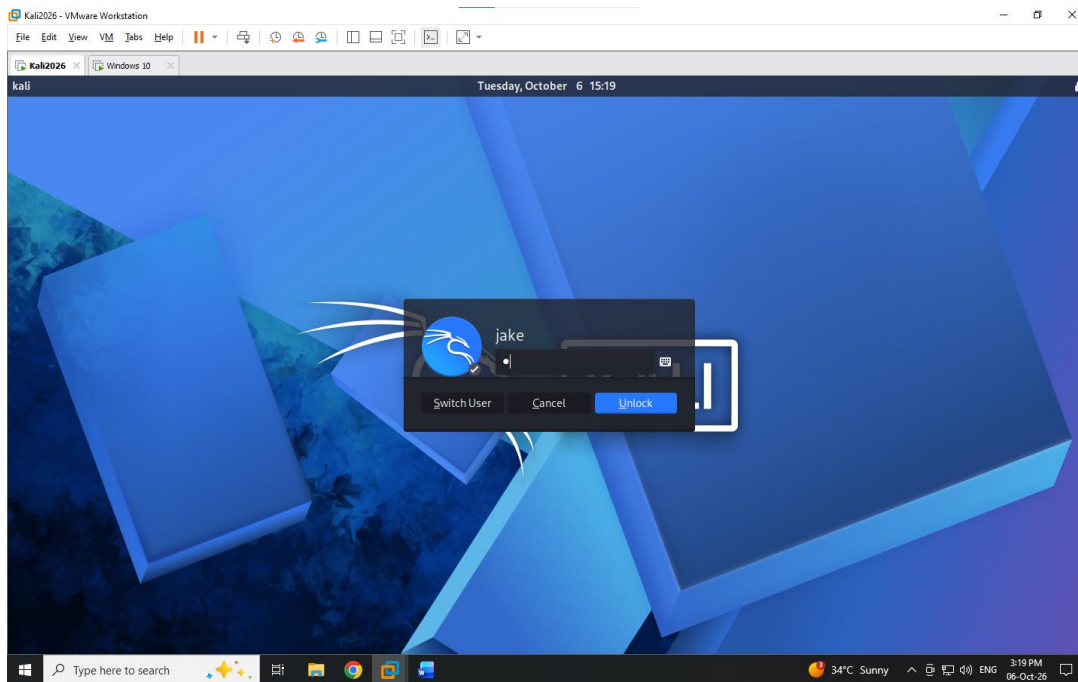


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Q.1 Install Ubuntu (or any Linux distribution) on your system using a virtual machine tool like VirtualBox, and take a screenshot of your successful login to the Linux terminal.

ANS :

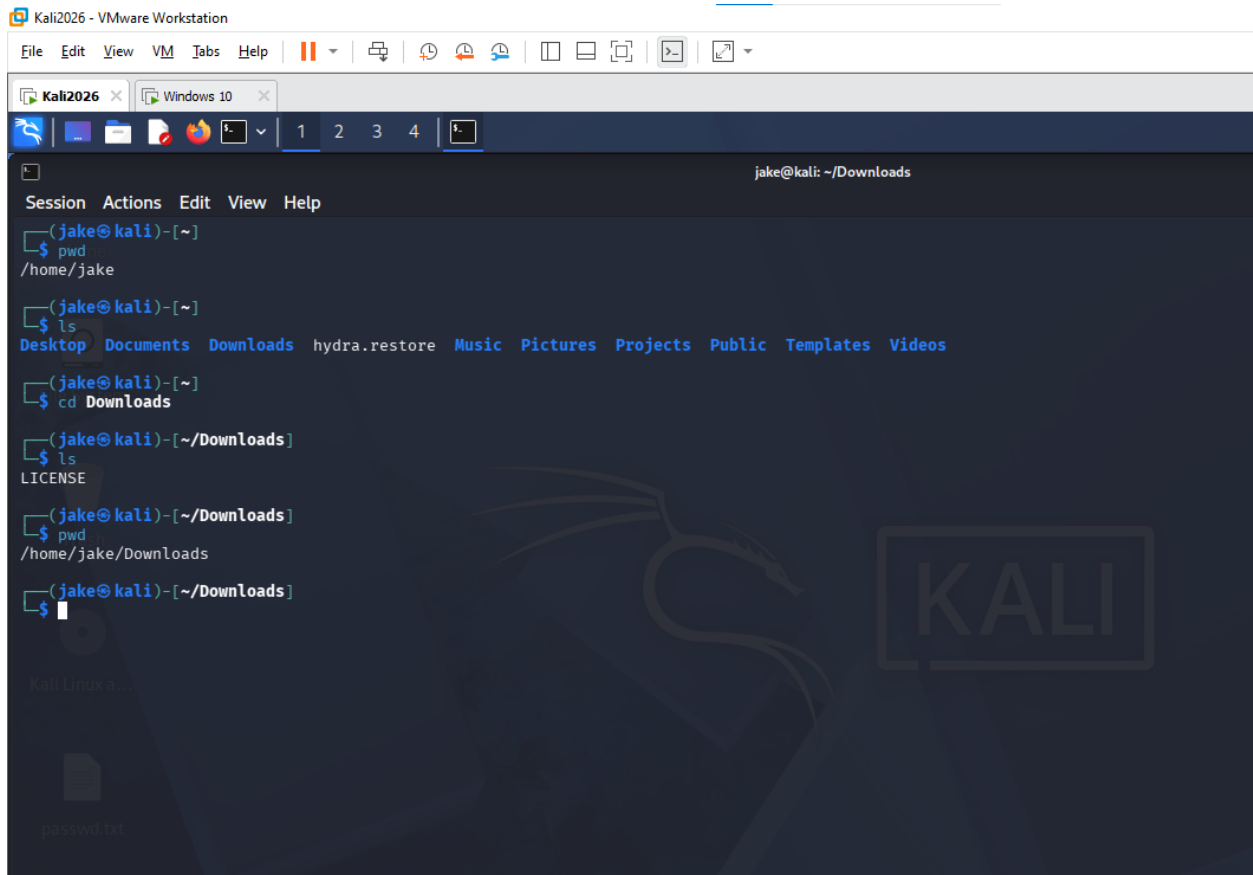


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Q.2 Open the Linux terminal and use the commands `pwd`, `ls`, and `cd` to navigate to your Downloads folder, then list all files present in it. Hint: Use '`cd ~/Downloads`' to move to the Downloads directory.

ANS :



The screenshot shows a Kali Linux terminal window within a VMware Workstation. The terminal displays the following sequence of commands and outputs:

```
jake@kali: ~/Downloads

Session Actions Edit View Help

(jake@kali)-[~]
$ pwd
/home/jake

(jake@kali)-[~]
$ ls
Desktop Documents Downloads hydra.restore Music Pictures Projects Public Templates Videos

(jake@kali)-[~]
$ cd Downloads

(jake@kali)-[~/Downloads]
$ ls
LICENSE

(jake@kali)-[~/Downloads]
$ pwd
/home/jake/Downloads

(jake@kali)-[~/Downloads]
$
```

The terminal background features a Kali Linux logo and the text "KALI".

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Q.3 Create a new directory named 'music_playlist' inside your home directory, and inside it, create three empty files named song1.mp3, song2.mp3, and song3.mp3 using a single Linux command.

ANS :

```
(jake@kali)-[~]
└─$ ls
Desktop  Documents  Downloads  hydra.restore  Music  Pictures  Projects  Public  song1.mp3  song2.mp3  song3.mp3  Templates  Videos

(jake@kali)-[~]
└─$ mkdir music_playlist ; touch music_playlist/song{1..3}.mp3

(jake@kali)-[~]
└─$ ls
Desktop  Documents  Downloads  hydra.restore  Music  music_playlist  Pictures  Projects  Public  song1.mp3  song2.mp3  song3.mp3  Templates  Videos

(jake@kali)-[~]
└─$ ls music_playlist
song1.mp3  song2.mp3  song3.mp3

(jake@kali)-[~]
└─$ ss
```

Q.4 Change the permissions of the 'music_playlist' directory so that only you (the owner) have read, write, and execute permissions, and no one else has any permissions. Hint: Use the chmod command with the correct numeric code.

ANS :

```
Kali2026 - VMware Workstation
File Edit View VM Tabs Help
Kali2026 x Windows 10 x
Session Actions Edit View Help

(jake@kali)-[~]
└─$ chmod 700 music_playlist

(jake@kali)-[~]
└─$ ls -l
total 52
drwxr-xr-x 2 jake jake 4096 Oct 6 09:26 Desktop
drwxr-xr-x 2 jake jake 4096 Jul 15 12:00 Documents
drwxr-xr-x 2 jake jake 4096 Sep 29 15:57 Downloads
-rw-rw-r-- 1 jake jake 10302 Oct 6 09:26 hydra.restore
drwxr-xr-x 2 jake jake 4096 Jul 15 12:00 Music
drwx----- 2 jake jake 4096 Oct 6 15:27 music_playlist
drwxr-xr-x 2 jake jake 4096 Jul 15 12:00 Pictures
drwxr-xr-x 2 jake jake 4096 Jul 15 12:00 Projects
drwxr-xr-x 2 jake jake 4096 Jul 15 12:00 Public
-rw-rw-r-- 1 jake jake 0 Oct 6 15:25 song1.mp3
-rw-rw-r-- 1 jake jake 0 Oct 6 15:25 song2.mp3
-rw-rw-r-- 1 jake jake 0 Oct 6 15:25 song3.mp3
drwxr-xr-x 2 jake jake 4096 Jul 15 12:00 Templates
drwxr-xr-x 2 jake jake 4096 Jul 15 12:00 Videos

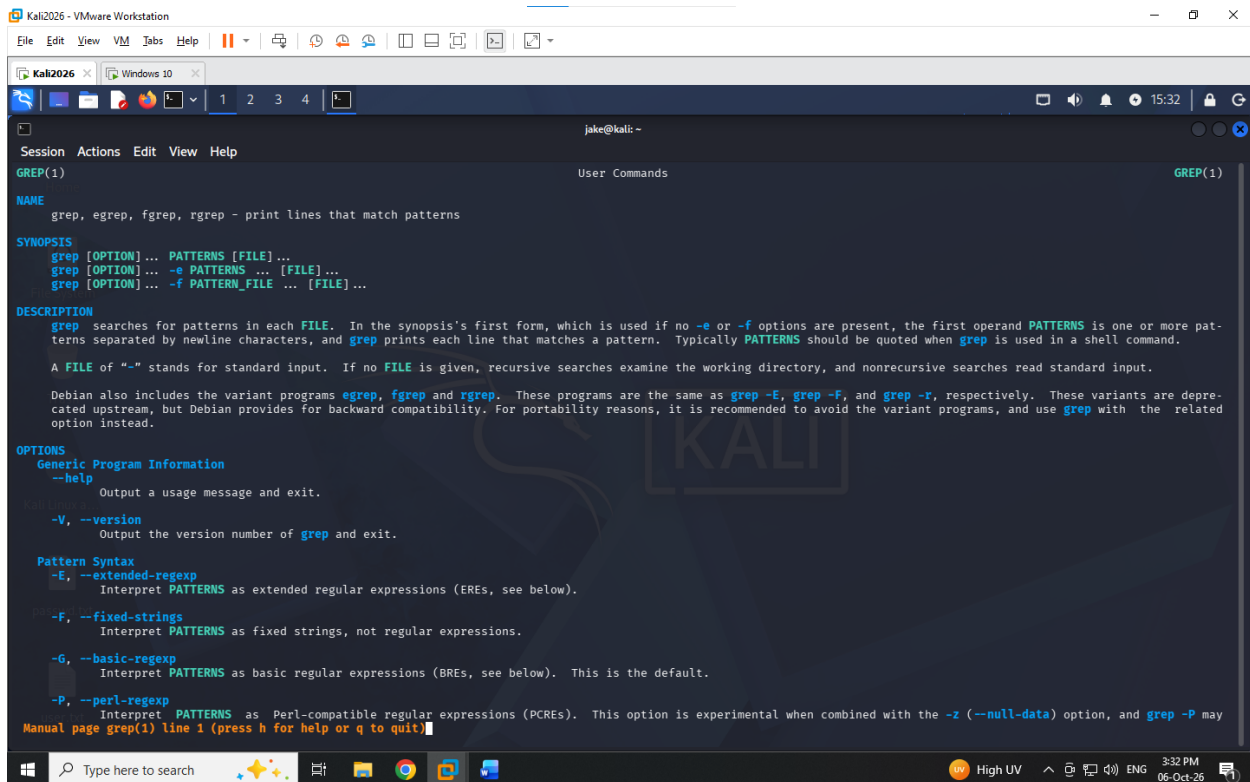
(jake@kali)-[~]
└─$
```

ASSIGNMENT : 7 – LINUX ESSENTIALS

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Q.5 Use the man command in the Linux terminal to find out what the 'grep' command does, and write a one-line summary in your own words.

ANS :



The screenshot shows a Kali Linux terminal window with the man page for the `grep` command. The terminal title is "Kali2026 - VMware Workstation". The terminal shows the command `grep(1)` and the resulting man page content. The man page includes sections for NAME, SYNOPSIS, DESCRIPTION, and OPTIONS. The DESCRIPTION section explains that `grep` searches for patterns in each file, and the OPTIONS section lists various flags like `--help`, `--version`, `--extended-regexp`, `--fixed-strings`, `--basic-regexp`, and `--perl-regexp`.

```
Kali2026 - VMware Workstation
File Edit View VM Tabs Help
Kali2026 x Windows 10 x
1 2 3 4
jake@kali: ~
Session Actions Edit View Help
GREP(1) User Commands GREP(1)
NAME
grep, egrep, fgrep, rgrep - print lines that match patterns
SYNOPSIS
grep [OPTION]... PATTERNS [FILE]...
grep [OPTION]... -e PATTERNS ... [FILE]...
grep [OPTION]... -f PATTERN_FILE ... [FILE]...
DESCRIPTION
grep searches for patterns in each FILE. In the synopsis's first form, which is used if no -e or -f options are present, the first operand PATTERNS is one or more pat-
terns separated by newline characters, and grep prints each line that matches a pattern. Typically PATTERNS should be quoted when grep is used in a shell command.
A FILE of "-" stands for standard input. If no FILE is given, recursive searches examine the working directory, and nonrecursive searches read standard input.
Debian also includes the variant programs egrep, fgrep and rgrep. These programs are the same as grep -E, grep -F, and grep -r, respectively. These variants are depre-
cated upstream, but Debian provides for backward compatibility. For portability reasons, it is recommended to avoid the variant programs, and use grep with the related
option instead.
OPTIONS
Generic Program Information
--help
Output a usage message and exit.
-V, --version
Output the version number of grep and exit.
Pattern Syntax
-E, --extended-regexp
Interpret PATTERNS as extended regular expressions (EREs, see below).
-F, --fixed-strings
Interpret PATTERNS as fixed strings, not regular expressions.
-G, --basic-regexp
Interpret PATTERNS as basic regular expressions (BRES, see below). This is the default.
-P, --perl-regexp
Interpret PATTERNS as Perl-compatible regular expressions (PCREs). This option is experimental when combined with the -z (--null-data) option, and grep -P may
Manual page grep(1) line 1 (press h for help or q to quit)
```

Grep command is used to search the given pattern

Grep command search for pattern in each file